Robust UAV–Satellite Imagery Alignment via Cross-Domain Descriptor Learning and Homography Optimization

Yang Xiao, Yunbo Wu, Chen Yang, Zhen Qin, Ziping Yu, and Huilong Zheng

***Abstract*—Localization methods that leverage satellite maps and their associated geospatial information are critical for uncrewed aerial vehicles (UAVs) executing time-sensitive missions in global navigation satellite system (GNSS)-denied environments. UAV and satellite images exhibit substantial cross-domain differences, including appearance variations caused by temporal changes and geometric discrepancies between perspective imaging and orthorectification. However, existing methods in the UAV–satellite setting often lack effective supervision for local feature matching, which limits the cross-domain adaptation capability of feature matching models. Existing datasets also provide limited support for rigorous fine-grained registration evaluation. To address these issues, we present Co-Lo, a calibrated UAV–satellite dataset. During dataset construction, dense matching is combined with sparse keypoints to obtain reliable keypoint correspondences. An initial calibration procedure is then used to derive accurate ground-truth homography matrices for evaluation. Based on these supervisory signals, we validate the effectiveness of the proposed weakly supervised cross-domain adaptation strategy. In addition, a dual evaluation protocol is designed to decouple algorithm robustness (across texture variations) from sequence stability (along flight trajectories). Compared with representative sparse-matching and semi-dense-matching baselines, the adapted models show improved robustness in cross-domain matching. In texture-sparse scenes, they also yield relatively stable continuous localization performance.**

***Index Terms*—Absolute visual localization, cross-domain matching, global navigation satellite system (GNSS)-denied, image registration, satellite imagery, uncrewed aerial vehicle (UAV).**

I. Introduction

UNCREWED Aerial Vehicles (UAVs) have become essential for aerial operations such as large-scale mapping [1], agricultural monitoring [2], and emergency response [3], [4]. While these platforms typically rely on integrated global navigation satellite system (GNSS)/INS systems for navigation, the susceptibility of GNSS to interference and spoofing remains a critical vulnerability [5]. Although visual odometry and SLAM [6], [7], [8] offer reliable solutions in GNSS-denied indoor or low-altitude environments, achieving stable, long-distance visual localization in open-sky scenarios remains challenging [9]. Consequently, visual localization methods that establish correspondences between UAV imagery and georeferenced maps have gained significant traction.

Reference maps typically consist of 3-D models [10], digital elevation models (DEMs) [11], or satellite imagery [12]. Methods leveraging 2-D satellite maps—which assume a nadir (downward-looking) UAV perspective and a planar ground surface—are particularly advantageous due to their rapid deployment, streamlined workflows, and lower costs. These factors are vital for time-critical missions. Accordingly, our work focuses on this satellite-based localization paradigm.

To minimize the discrepancy between UAV and satellite images, UAVs typically capture images from a nadir view. In this pipeline, the UAV is initialized with precise coordinates, and subsequent continuous visual localization (CVL) is maintained by establishing pixel-to-coordinate mappings [13] or estimating homography matrices (H) [14].

However, further progress in UAV–satellite image registration and alignment remains constrained by two fundamental issues. First, existing learning-based local feature matching methods are typically developed using training data with explicit geometric correspondences, such as supervision derived from camera poses, depth information, or multiview reconstruction. Such supervision provides strong pixel-level constraints for local feature learning. In UAV–satellite registration, however, explicit camera correspondences are generally unavailable, which limits the availability of cross-domain supervision for training local matching models. Second, existing UAV–satellite datasets often exhibit varying degrees of geospatial misalignment among UAV images, GNSS ground truth, and satellite images. This geospatial inconsistency not only affects the evaluation of fine-grained registration accuracy but may also obscure the actual performance gains brought by cross-domain adaptation methods.

To address these issues, this work investigates how dense correspondences produced by dense matchers can be converted into trainable supervisory signals in the absence of explicit camera correspondences and pixel-level annotations. We introduce a supervision transfer strategy that converts

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dense cross-domain correspondences into pseudo-supervision for local descriptor learning. In addition, these supervisory signals can be further transformed into homography matrices, which are then used to supervise a deep Lucas–Kanade (DLK [15]) optimizer, enabling its adaptation to low-texture UAV–satellite scenarios.

On this basis, we construct the Co-Lo dataset. In this dataset, initial homography matrices are refined using manually annotated ground control points (GCPs). Co-Lo supports a more rigorous evaluation of fine-grained registration accuracy. For example, for images captured at an altitude of 400 m, the mapping error can be controlled within 3 pixels (i.e., <1 m).

We further introduce a dual evaluation protocol. The first component is algorithmic evaluation, which measures the registration quality of individual UAV–satellite pairs. The second component is sequential evaluation, which measures the projected localization error along flight trajectories and the stability of CVL.

In summary, the primary contributions of this work are as follows.

1) We present Co-Lo, a calibrated UAV–satellite dataset containing challenging image pairs and real flight sequences, which supports the evaluation of registration accuracy and sequential localization stability.
2) We introduce a weakly supervised cross-domain adaptation strategy that converts dense correspondences into pseudo-labels for local descriptor learning on repeatable keypoints, thereby enabling descriptor adaptation without manual annotation.
3) We further show that pseudo-homography matrices can serve as effective supervision for deep Lucas–Kanade (DLK) optimization, facilitating UAV–satellite image alignment in texture-sparse environments.

II. Related Work

Some approaches [16], [17], [18] rely on preconstructed 3-D models and estimate camera poses via the perspective-n-point (PnP) algorithm combined with feature matching. However, these methods entail high computational costs and have limited spatial scalability. Other works [11], [19] incorporate digital surface models (DSMs) for PnP-based localization; yet, the scarcity and high acquisition cost of high-precision elevation data limit the generalizability of these solutions. Consequently, our work focuses on 2-D satellite map-based methods.

We review 2-D satellite map-based approaches, including image retrieval-based and image registration-based visual localization methods, and analyze their primary limitations.

A. Visual Localization Methods

Retrieval-based cross-view geo-localization methods [20], [21], [22], [23] employ global descriptors, such as NetVLAD [24], to retrieve the satellite tile from a prebuilt database that most closely matches the current aerial perspective. The database typically partitions the target area into a discrete grid, using tile centers as positional labels. While computationally efficient, these methods provide only coarse localization, with precision constrained by the spatial resolution of the tiles.

Matching/registration-based refinement methods [25], [26], [27], [28] typically identify a candidate region via global retrieval or odometry, followed by pixel-level alignment within the candidate region using local feature matching (e.g., SuperPoint [29] + SuperGlue [30]). However, effective supervision for local feature matching remains limited in the UAV–satellite setting. Descriptor adaptation is challenging because most learning-based local feature methods rely on explicit correspondences. As a result, existing studies usually validate localization methods by directly using pretrained models in texture-rich urban environments. This may lead to performance degradation in low-texture scenes, such as farmlands, grasslands, and deserts. Although [31] adopts a self-training boosted matching method, this strategy mainly optimizes descriptors associated with existing keypoints and is difficult to extend to more general keypoints. Therefore, we introduce a weakly supervised cross-domain adaptation strategy that learns repeatable keypoint descriptors from dense correspondences, rather than relying on the model's own correspondences.

DLK optimization has also been explored to alleviate the limitations of explicit keypoint matching. Some works [14], [32], [33] use the DLK framework to iteratively optimize homography parameters by minimizing feature-based reprojection errors. However, due to limitations in available data, these methods are typically trained only on prealigned image pairs. Such a training scheme is difficult to apply to UAV–satellite image pairs with perspective variation. We further convert dense correspondences into homography matrices to generate supervisory signals for image pairs, thereby supporting the cross-domain adaptation of the DLK model.

B. Visual Localization Datasets

Cross-view retrieval datasets, such as University-1652 [34] and SUES-200 [35], are widely used to benchmark the performance of global descriptors in aerial/ground-to-satellite matching. However, these datasets typically consist of discretely sampled oblique-view imagery and are not well suited to evaluating continuous localization over flight trajectories. Other datasets, such as ALTO [36] and VPAIR [37], provide nadir UAV–satellite pairs with high visual overlap. While continuous localization can be approximated in these datasets through image retrieval within a candidate range, the localization error is intrinsically bounded by the sampling density and overlap ratio of the reference images. For instance, DenseUAV [21] uses an 80% overlap between adjacent frames. While this improves retrieval performance, it introduces substantial spatial redundancy and focuses on retrieval rather than the efficacy of the matching algorithms themselves.

A critical limitation of current datasets, such as UAV-VisLoc [38] and AerialVL [39], is the geospatial misalignment among the UAV viewpoint center, satellite images, and onboard GNSS records (as shown in Table I). These discrepancies often necessitate the use of overly permissive success thresholds; for example, AerialVL employs a 50 m radius to define a "successful" localization. While acceptable for coarse georegistration, such thresholds are inadequate for fine-grained localization where rigorous error metrics are required. At the same time, larger error tolerances may make it difficult

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TABLE I
POSITIONAL DISCREPANCIES BETWEEN CAMERA CENTER PROJECTIONS AND GNSS GROUND-TRUTH. IR: IMAGE RETRIEVAL. IM: IMAGE MATCHING

Dataset	Method	Error (m)		Sample Size
		Mean	Max	
AerialVL	IR & IM	25.49	55.82	537
UAV-VisLoc	IM	27.13	88.23	298
VPAIR	IR	12.82	30.78	332

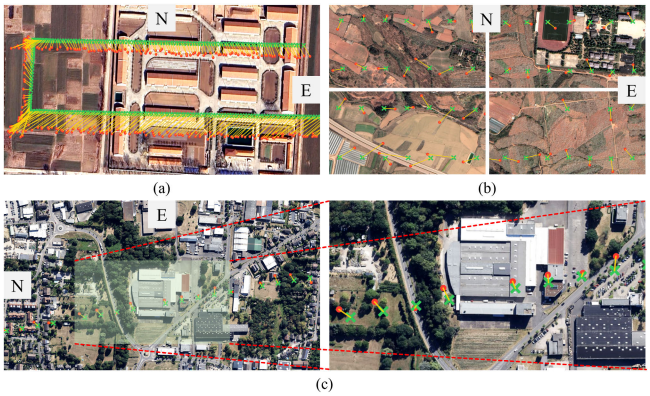


Fig. 1. Error between the projected image-center location and the onboard GNSS. (a)–(c) Error examples of AerialVL, UAV-VisLoc, and VPAIR datasets, respectively.

to quantify the performance gains brought by cross-domain adaptation.

As illustrated in Fig. 1, the green "×" represents the GNSS coordinate, while the red "·" indicates the ground projection of the camera center. This offset is primarily caused by nonnadir flight attitudes (pitch and roll), which shift the camera's optical center away from the nadir GNSS position. (Fig. 1(c) has been reoriented for improved visualization; please refer to the color version for clarity).

Existing datasets either provide reference imagery with limited environmental diversity or lack the ground-truth necessary to evaluate the quantitative contribution of image matching to the localization pipeline. Consequently, prior work is confined to performance comparisons within specific, narrow scenarios.

To address these gaps, we introduce the Co-Lo dataset. We evaluate model generalization using discretely sampled UAV–satellite pairs and assess continuous localization performance using real-world flight sequences across large-scale satellite maps. Furthermore, we shift the evaluation paradigm from image retrieval success rates to matching-based metrics, namely mean matching accuracy (MMA) and mean homography accuracy (MHA). Using the homography matrices provided in Co-Lo to calibrate geospatial offsets, we enable evaluation under stricter localization thresholds of 2–10 m.

III. CO-LO DATASET

In this section, we first provide an overview of the proposed dataset's characteristics. We then describe the construction of UAV–satellite image matching pairs under conditions of significant cross-domain discrepancy.

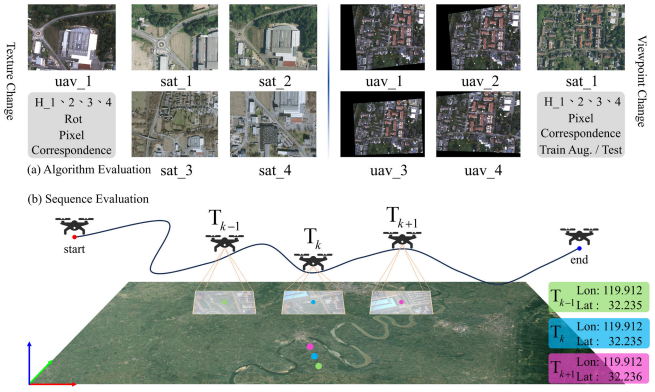


Fig. 2. Example samples for (a) algorithmic evaluation and (b) sequential evaluation. (a) Multitemporal satellite References and synthetic homography perturbations. (b) Dynamic window cropping centered on the previous prediction.

A. Dataset Characteristics

To facilitate practical application, our dataset focuses on near-planar homography-based correspondence rather than 6-DoF pose recovery. Fig. 2 illustrates the core components of the dataset.

The Co-Lo dataset is divided into two tasks: algorithmic evaluation and sequential evaluation. For the former, we collect UAV images from public repositories and synthetic data generated through realistic imaging pipelines. These samples span diverse geomorphological landscapes and altitudes. By selecting multitemporal and multiscale satellite reference maps, we explicitly model challenges such as seasonal transitions, varying lighting conditions, and land changes. The dataset comprises 1000 unique UAV images, split evenly between texture-rich and texture-sparse environments, with an 800/100/100 ratio for training, validation, and testing, respectively. Each UAV image forms four distinct matching pairs.

For sequential evaluation, we use continuous real-world flight trajectories to measure localization accuracy. In this mode, satellite imagery is selected to be as close in time as possible to the UAV flight date to reflect operational realities.

B. Dataset Construction

UAV Data Sampling: The source data for algorithmic evaluation comprises VPAIR and UAV-VisLoc. These sources provide diverse environments and scenarios. For sequential sampling, we use real-world flight trajectories from AerialVL and DenseUAV, applying rigorous alignment corrections to resolve their inherent geospatial offsets.

Satellite Data Sampling: To accommodate varying UAV flight altitudes, we collected satellite imagery across zoom levels 18–21. Furthermore, to evaluate matching robustness under different textural and seasonal conditions, we gathered multitemporal imagery spanning from 2017 to 2025. This temporal diversity introduces significant variations in land cover, vegetation, and lighting.

Data Processing and Calibration: To establish reliable correspondences under extreme cross-domain appearance gaps, we use RoMa [40]—a dense matcher that fuses DINOv2 semantic features with finegrained local features. RoMa produces

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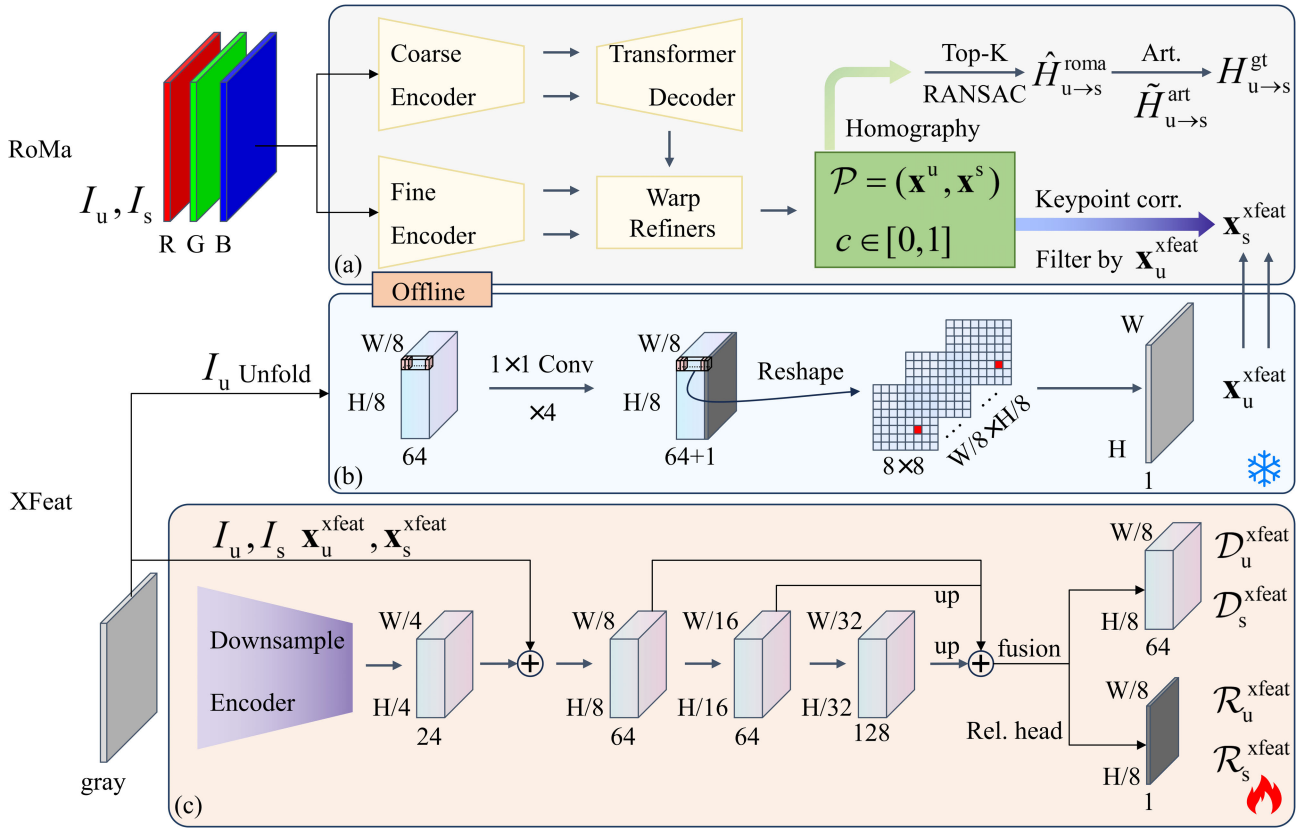


Fig. 3. Data processing and descriptor training. (a) Acquisition of the ground-truth homography $H^{gt}_{u\to s}$ for I_u and I_s, as well as the sampled correspondence set $\mathcal{P}$. (b) Keypoint module, which remains frozen during training. (c) Training pipeline for descriptors and reliability maps. Keypoint corr.: Keypoint correspondence and Rel. head: Reliability head.

pixel-to-pixel correspondences even under substantial texture discrepancies. For UAV image I_u and satellite image I_s, we sample correspondences from RoMa's dense correspondence field, denoted as $\mathcal{P} = \{(x^u_0, x^s_0), (x^u_1, x^s_1), \ldots, (x^u_{19999}, x^s_{19999})\}$, to establish initial geometric mappings $\hat{H}^{roma}_{u\to s}$, and then perform manual calibration to refine the homography $\Delta H^{art}_{u\to s}$. This procedure simultaneously achieves precise alignment between the UAV camera-center projection and the satellite image. The ground-truth homography $H^{gt}_{u\to s}$ can be formally represented as:

$$\begin{aligned} \mathcal{P} &= \text{Sample}\left(\text{RoMa}\left(I_u, I_s\right)\right) \\ \hat{H}^{roma}_{u\to s} &= \text{RANSAC}\left(\mathcal{P}\right) \\ H^{gt}_{u\to s} &= \Delta H^{art}_{u\to\ s} \hat{H}^{roma}_{u\to s}\ . \end{aligned} \tag{1}$$

The overall workflow is illustrated in Fig. 3(a), and a detailed description is given in Appendix A.

In the absence of pixel correspondences derived from camera geometry, we adopt a weakly supervised cross-domain adaptation strategy to convert dense correspondences into sparse and repeatable point pairs. To establish reliable keypoint correspondences between I_u and I_s, we detect a set of keypoints on the UAV image using XFeat [41], denoted as $\mathcal{M} = \{\mathbf{x}^u_0, \mathbf{x}^u_1, \ldots, \mathbf{x}^u_{4095}\}$. We use RoMa to obtain each pair (x^u_j, x^s_j) in the sampled correspondence set $\mathcal{P}$ with confidence score c_j, and discard pairs with $c_i < \tau$ to yield $\mathcal{P}_\tau$. For each keypoint $\mathbf{x}^u_i$ in $\mathcal{M}$, we find its nearest neighbor x^u_j in $\mathcal{P}_\tau$ and take the paired satellite point x^s_j as the match. As this procedure is computationally intensive, it is conducted offline. The process is illustrated in Fig. 3(a) and (b).

IV. Evaluation Metrics

We define the metrics for two distinct application scenarios: algorithmic evaluation and sequential evaluation.

A. Algorithmic Evaluation

We provide ground-truth homography matrices in the Co-Lo dataset for this evaluation. We employ two metrics from the image matching domain [42]: MMA and MHA, defined as follows:

$$\begin{aligned} \text{MMA@}\tau &= \frac{1}{N}\sum_{i=1}^{N} 1\left[p_i \le \tau\right] \\ \text{MHA@}\tau &= \frac{1}{N}\sum_{i=1}^{N} 1\left[c_i \le \tau\right] \end{aligned} \tag{2}$$

where p_i denotes the pixel-wise matching error, and c_i represents the average reprojection error of the four image corners. τ is the error threshold. We perform quantitative analysis across multiple thresholds, $\tau \in \{3, 5, 7, 10\}$ px. MMA measures correspondence accuracy, whereas MHA evaluates global alignment. Because correct inliers may still cluster locally, MHA complements MMA by assessing wholeimage registration.

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B. Sequential Evaluation

We quantify localization error by the geographic distance between the latitude–longitude of the image-center projection on the satellite map and the corresponding ground-truth latitude–longitude provided by Co-Lo. Mean absolute error (MAE) and root mean square error (RMSE) are introduced as primary metrics:

$$\text{MAE} = \frac{1}{N}\sum_{i=1}^{N}|\hat{y}_i - y_i|$$
$$\text{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}(\hat{y}_i - y_i)^2} \tag{3}$$

where $|\hat{y}_i - y_i|$ represents the WGS-84 geodesic distance error in meters.

To assess the reliability of the localization pipeline, we propose two additional metrics: the maximum continuous stable localization ratio ($\text{R}_{\max}$) and the successful coverage ratio (R_{suc}). $\text{R}_{\max}$ measures the longest continuous segment of successful localization, while R_{suc} evaluates the overall stability throughout the trajectory. A "successful localization" is defined by a distance error of less than 10 m. $\text{R}_{\max}$ and R_{suc} are expressed as:

$$\text{R}_{\max} = \frac{\sum_{s=1}^{S} L_s^{\max}}{\sum_{s=1}^{S} D_s}, \text{R}_{\text{suc}} = \frac{\sum_{s=1}^{S} L_s^{\text{sum}}}{\sum_{s=1}^{S} D_s} \tag{4}$$

where $L_s^{\max}$, L_s^{sum}, and D_s represent the maximum continuous successful localization distance, the total successful distance, and the total trajectory distance for a given sequence, respectively. S denotes the total number of sequences. Detailed definitions of the metrics are given in Appendix B.

V. Baseline Models

To improve descriptor adaptability to cross-domain imagery and address the challenges of texture-sparse environments, we perform enhanced training on the Co-Lo dataset. This includes an XFeat-based descriptor enhancement model and a DLK-based iterative homography optimization model.

A. XFeat-Based Enhancement

We use the lightweight XFeat [41] as our feature extraction backbone. The model comprises three primary modules: keypoint detection, descriptor extraction, and reliability map estimation [as illustrated in Fig. 3(b) and (c)]. The efficiency of the keypoint module is achieved through a teacher-student knowledge distillation framework.

The original XFeat is trained on MegaDepth [43], where reprojections are derived from camera poses and depth maps. Constructing such reprojections is highly challenging in UAV–satellite scenarios. In this article, the point correspondences provided by Co-Lo (see Section III-B) are used to optimize the descriptor and reliability head, so as to obtain cross-domain adaptive feature representations (XFeat*). During this weakly supervised process, the keypoint detection module [Fig. 3(b)] is kept frozen.

Let $\mathcal{D}_1$ and $\mathcal{D}_2$ denote the 64-D descriptors extracted from the UAV and satellite image patches, respectively. $\mathcal{D}_1(i,\cdot)$ and $\mathcal{D}_2(i,\cdot)$ denote the descriptor correspondences established from the point correspondences provided by Co-Lo. The descriptor loss, $\mathcal{L}_{ds}$, is formulated using a dual-softmax operator on the similarity matrix $\mathbf{S}^{N\times N} = \mathcal{D}_1\mathcal{D}_2^{\text{T}}$, where the diagonal elements represent the similarities of corresponding pairs $\mathcal{D}_1(i,\cdot)$ and $\mathcal{D}_2(i,\cdot)$

$$\mathcal{L}_{ds} = -\sum_i \log\left(\text{softmax}\left(\mathbf{S}_{ii}\right)\right) - \sum_i \log\left(\text{softmax}\left(\mathbf{S}^{\text{T}}\right)_{ii}\right) \tag{5}$$

where $\mathbf{S}_{ii}$ denotes the elements along the main diagonal of the similarity matrix $\mathbf{S}$.

The probability of a correct match is inferred from the product of the maximum probabilities in each row of the similarity matrix $\mathbf{S}$. As descriptors become more similar along the diagonal, the matching probability increases. Thus, the reliability maps are optimized indirectly. Consistent with the original XFeat implementation, the reliability loss $\mathcal{L}_{\text{rel}}$ is defined as:

$$\mathcal{L}_{rel} = \left|\sigma\left(\mathbf{R}_1\right) - \overline{\mathbf{R}}_1 \odot \overline{\mathbf{R}}_2\right| + \left|\sigma\left(\mathbf{R}_2\right) - \overline{\mathbf{R}}_1 \odot \overline{\mathbf{R}}_2\right| \tag{6}$$

where $\sigma(\cdot)$ denotes the sigmoid function; $\overline{\mathbf{R}}_1 = \max_{\text{r}}(\text{softmax}_{\text{r}}(\mathbf{S})), \overline{\mathbf{R}}_2 = \max_{\text{r}}(\text{softmax}_{\text{r}}(\mathbf{S}^{\text{T}}))$; r denotes the row operation; $\odot$ denotes dot product.

The final objective function is a combination of the descriptor loss and the reliability loss:

$$\mathcal{L} = \mathcal{L}_{ds} + \mathcal{L}_{rel}. \tag{7}$$

Implementation Details: We initialized the model with pretrained XFeat weights and froze the keypoint detection branch. The MegaDepth training pipeline was replaced with the Co-Lo UAV–satellite pairs. For self-supervised augmentation, we applied synthetic homography transformations to the COCO20k dataset. We used the Adam optimizer with a learning rate of 5×10^{-5}, while other hyperparameters remained consistent with [41]. Training was conducted on a single NVIDIA RTX 3090 GPU for 160000 steps, and the model reached convergence in approximately two days.

B. DLK-Based Homography Optimization

Our implementation of the DLK framework is based on [14]. Previous DLK models were trained exclusively on multitemporal satellite image pairs and therefore fail to generalize well to real UAV–satellite image pairs. To address this limitation and meet the requirements of texture-sparse localization, we retrained DLK on Co-Lo to evaluate the effect of using real UAV–satellite training pairs.

To enhance iterative efficiency, we employ the Inverse Compositional Lucas–Kanade (IC-LK) algorithm [15], [32], [33], with details given in Appendix C. At iteration k, given the current parameters $\mathbf{p}_k$, the goal is to estimate an incremental update $\Delta\mathbf{p}$ to align the input image $I(x)$ with the template image $T(x)$ by minimizing the following energy function $E(\Delta\mathbf{p})$:

$$E\left(\Delta\mathbf{p}\right) = \sum_{\mathbf{x}} \left\| T\left(W\left(\mathbf{x};\Delta\mathbf{p}\right)\right) - I\left(W\left(\mathbf{x};\mathbf{p}_k\right)\right)\right\|_2^2. \tag{8}$$

Unlike classical LK, which minimizes photometric error in the image-intensity domain, DLK optimizes alignment in feature space using CNN feature maps. This often improves

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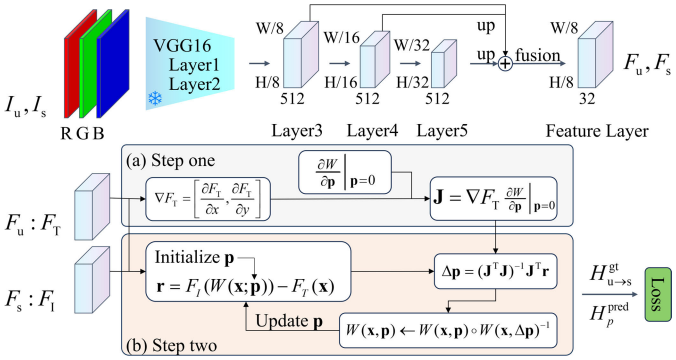


Fig. 4. Schematic of the DLK model. The first two VGG16 blocks are frozen, and features from the last three blocks are fused in an XFeat-style manner. In IC-LK, the satellite image and the UAV image are designated as the input and the template, respectively, which avoids inverting $H^{\mathrm{gt}}_{\mathrm{u}\to\mathrm{s}}$. (a) Step one. (b) Step two.

robustness to texture and illumination variations. Thus, for feature maps F_I and F_T, $E(\Delta\mathbf{p})$ can be written as:

$$E(\Delta\mathbf{p}) = \sum_{\mathbf{x}} \left\| F_I\left(W\left(\mathbf{x};\mathbf{p}_k\right)\right) - F_T\left(W\left(\mathbf{x};\Delta\mathbf{p}\right)\right)\right\|_2^2. \quad (9)$$

The overall architecture (DLK*) is illustrated in Fig. 4. We introduce the following modifications to the original DLK [14].

1) We use a pretrained VGG16 [44] architecture and fuse multiscale features from the last three layers. The fused feature map is down-sampled to W/8 × H/8 × 32, to reduce computational overhead.
2) We replace the original 6-parameter affine with a full 8-DoF homography to better account for camera tilt and perspective effects.

The model (DLK*) was trained by minimizing the 4-point corner

$$\mathrm{L} = \sum_{j=1}^{4} \left\| W\left(\mathbf{e}_j;\mathbf{p}_1\right) - W\left(\mathbf{e}_j;\mathbf{p}_2\right)\right\|_2^2 \quad (10)$$

where $\mathbf{p}_1$ and $\mathbf{p}_2$ represent the ground-truth and predicted corner coordinates, respectively.

We used 800 UAV images from Co-Lo for training. Each UAV image was paired with four satellite references, yielding 3200 training pairs. To improve generalization, random homography perturbations were applied at each epoch. All input image pairs were resized to 640 × 640. Training was performed on a single NVIDIA RTX 3090 GPU using the same hyperparameter settings as in [14], and the model converged in approximately 1.5 days.

VI. Experiments

We evaluate the adapted models and analyze their performance. Section VI-A describes the implementation details and experimental setup. In Section VI-B, we perform a comparative analysis of XFeat* and DLK* against several state-of-the-art (SOTA) baselines. Section VI-C evaluates the effectiveness of the ground-truth homography matrices and cross-domain point pairs provided by the Co-Lo dataset. Finally, Section VI-D discusses the limitations of our approach.

A. Implementation Details

We used XFeat* as the feature extractor, paired with LightGlue (LG) [45] for feature matching. For DLK*, an initial transformation $\mathbf{p}_0$ [32] was required. In texture-rich environments, $\mathbf{H}_0$ was estimated by SuperPoint [29] + SuperGlue [30] (SP + SG) and then converted to $\mathbf{p}_0$. In texture-sparse scenarios, where feature-based initialization is unreliable, $\mathbf{p}_0$ was initialized by applying random Gaussian noise to the four corner coordinates of the ground-truth homography projection on the satellite image (with standard deviation σ = 10 px).

For matching-based methods, all input image pairs were resized to 800 × 600. For DLK-based methods, images were resized to 640 × 640. For detector-based matchers, the keypoint detection threshold was set to 0.005, with a maximum of 4096 keypoints, and the matching confidence threshold was set to 0.2. Homography estimation was performed using OpenCV's USAC_MAGSAC with a 3-pixel reprojection threshold, confidence 0.9999, and a maximum of 10000 iterations.

B. Performance Evaluation

The performance of the cross-domain adapted models (XFeat* and DLK*) was evaluated on the Co-Lo test set (comprising 400 UAV–satellite image pairs, equally split between texture-rich and texture-sparse scenes) using the metrics established in Section IV. We mainly focused on the limitations of zero-shot methods and the gains brought by cross-domain adaptation. Inference was conducted within the image-matching-webui framework. We compared our methods against several representative baselines: SIFT [46] (a traditional local feature method); SP + SG, ALIKED [47] + LG, and XFeat + LG (detector-based methods); and LoFTR [48] and ASpanFormer [49] (detector-free methods).

1) Algorithmic Evaluation: According to the results in Table II, XFeat* achieves an MMA@10 of 77.1%, which is 0.7 pp higher than that of SP + SG, and an MHA@10 of 58.0%, which is 19.2 pp higher. The number of inliers for XFeat* reaches 276, exceeding SP + SG by 124. DLK* achieves an MHA@10 of 68.3%, which is 29.5 pp higher than that of SP + SG. Compared with zero-shot methods, the adapted methods yield higher performance. In particular, DLK* performs better in texture-sparse scenes. One possible explanation is that deep features capture geometric cues (e.g., roads and field boundaries) that remain informative even when salient keypoints are scarce. It is worth noting that the gap between XFeat* and SP + SG is small in terms of MMA, whereas the improvement is more pronounced in terms of MHA. This suggests that, in texture-sparse environments, zero-shot methods may still establish some correct inlier correspondences in local regions. However, if the inliers are unevenly distributed, the image pair may still fail to achieve correct global alignment.

Furthermore, the original XFeat achieves an MMA@10 of 65.9%, an MHA@10 of 22.5%, and 26 inliers, while XFeat* improves these values by 11.2 pp, 35.5 pp, and 250, respectively. Compared with the original method without cross-domain adaptation, the adapted version improves the

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TABLE II

COMPARISON OF MATCHING AND REGISTRATION METHODS ON TEXTURE-SPARSE SCENES IN CO-LO. RESULTS ARE REPORTED IN MMA AND MHA AT THRESHOLDS @3/@5/@7/@10 (HIGHER IS BETTER). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	Pub.'Year	MMA				MHA				Inliers
		@3	@5	@7	@10	@3	@5	@7	@10	
Traditional method										
SIFT	IJCV'04	4.2	5.3	5.7	6.2	–	–	–	–	–
Deep-learning detector-based methods										
SP + SG	CVPR'20	47.9	65.4	72.8	76.4	10.6	21.9	28.1	38.8	124
ALIKED + LG	TIM'23	39.9	54.4	58.5	61.3	10.0	20.6	27.5	35.0	97
XFeat + LG	CVPR'24	33.2	50.8	59.3	65.9	3.1	6.9	13.8	22.5	26
Deep-learning detector-free methods										
LoFTR	CVPR'21	39.7	52.5	56.8	60.0	7.5	17.5	24.4	33.8	54
ASpanFormer	ECCV'22	45.4	58.8	62.7	65.2	8.7	17.5	28.1	37.5	101
Ours										
XFeat* + LG	–	**48.8**	**67.4**	**73.5**	**77.1**	25.3	39.8	47.5	58.0	276
DLK*	–	–	–	–	–	**41.1**	**50.3**	**60.5**	**68.3**	–

TABLE III

COMPARISON OF CONTINUOUS LOCALIZATION METHODS IN CO-LO (AERIALVL) DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	MAE	RMSE	R_{max}	R_{suc}	Inliers
Detector-based methods					
SP + SG	0.79	1.49	71.2	95.6	84
ALIKED + LG	0.68	1.26	64.0	95.0	112
XFeat + LG	0.86	1.51	50.7	95.2	124
Detector-free methods					
LoFTR	0.83	1.64	51.0	95.1	190
ASpanFormer	0.65	1.30	51.0	96.2	339
Dense matching methods					
RoMa	**0.46**	**0.98**	78.7	**99.0**	**2184**
Ours					
XFeat* + LG	0.62	1.27	73.4	97.5	291
DLK*	2.87	2.90	**86.2**	98.7	–

number of point correspondences, matching accuracy, and global alignment quality. In addition, the original DLK does not produce effective results on Co-Lo and is therefore not included in Table II. This may be because its training data are mainly drawn from satellite image pairs and do not account for the domain gap between UAV and satellite images.

These results indicate that the proposed method establishes a larger number of reliable correspondences across cross-domain image pairs and improves overall image alignment. They also support the feasibility of using pseudo-supervision generated from dense correspondences for cross-domain training. Experimental results in texture-rich scenes are provided in Appendix D.

2) Sequence Evaluation: Table III summarizes the experimental results on the Co-Lo (AerialVL) sequences under a localization error threshold of 10 m. All matching-based methods achieve submeter mean localization error. This is primarily attributable to the calibration procedure, which eliminates the approximately 25-m georeferencing bias present in the original data, thereby providing a more objective measure of actual localization accuracy.

XFeat* achieves an MAE of 0.62 m, reducing the error by 0.06 m compared with ALIKED + LG. It attains R_{max} of 73.4%, which is 2.2 pp higher than SP + SG, and R_{suc} of 97.5%, which is 1.9 pp higher than SP + SG. The number of inliers reaches 291, which is 179 higher than that of ALIKED + LG. DLK* achieves R_{max} of 86.2%, which is 7.5 pp higher than RoMa. Compared with the original XFeat, XFeat* reduces the MAE by 0.24 m, improves R_{suc} by 2.3 pp, and increases the number of inliers by 167. The adapted methods achieve better performance, which is consistent with the observations in Section VI-B1. These results further support the effectiveness of cross-domain adaptation. Additional experimental results are given in Appendix E.

We note that DLK* yields a substantially higher MAE than the other matching-based methods. This may be related to several factors. DLK is essentially an iterative optimization method combined with learned features, and its optimization process remains sensitive to local optima [50]. In addition, the reduced resolution of the feature maps may lead to a loss of spatial detail. When the initialization is not sufficiently accurate, the model may converge to a nearby local solution or even fail to produce a valid result [32], [51]. Moreover, the large perspective differences between UAV and satellite images may further increase the difficulty of alignment.

3) Qualitative Analysis: To facilitate qualitative analysis, Fig. 5 presents visualizations of the matches produced by different methods. We select four scenes with progressively increasing difficulty, arranged from top to bottom. In texture-rich scenes, all methods are able to produce correct matches. As the scene gradually transitions to suburban, rural, and

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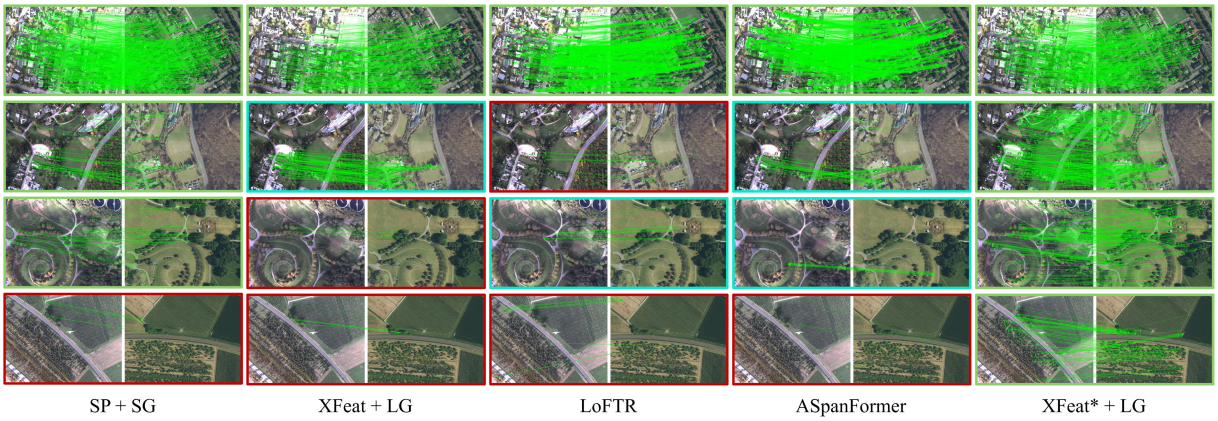


Fig. 5. Visual comparison of feature matching under varying texture conditions. Green boxes indicate success; blue boxes denote misalignment; red boxes indicate matching failure. Success/misalignment/failure are determined by the mean reprojection error over the annotated GCPs: < 5 px, 5–50 px, and > 50 px, respectively.

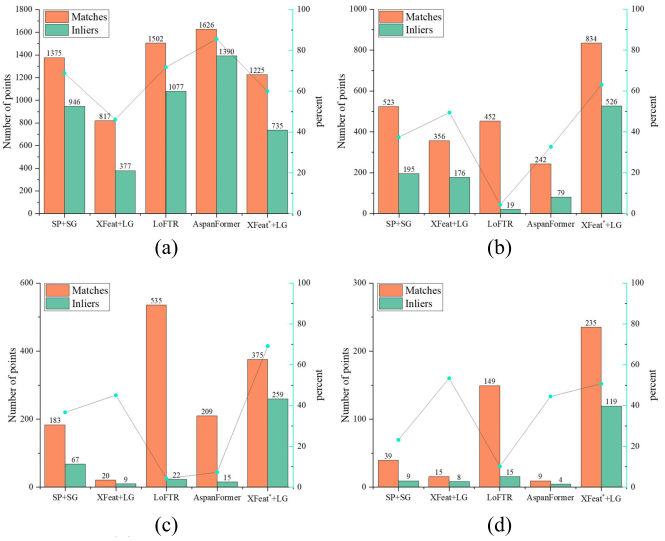


Fig. 6. Comparison of matching points and inlier counts across methods. Green dots represent the inlier-to-match ratio. (a) Scene 1. (a) Scene 2. (a) Scene 3. (a) Scene 4.

farmland areas, the baseline methods progressively fail because the number of reliable inliers becomes insufficient. In contrast, XFeat* still produces correct matches in these scenes. We compute inlier-to-match ratios for each method, as shown in Fig. 6. The results show that the inlier matching ratio of XFeat* remains above 50% across all four scenes, while the other methods exhibit less stable performance.

We further evaluate the methods using synthetically warped images. Fig. 7 presents visualizations, where the texture density decreases from top to bottom. The results show that, as texture becomes sparser, other methods may fail to register the images in challenging scenes, whereas DLK* and XFeat* remain stable under varying texture conditions. However, when the texture becomes even sparser, such as in the texture-sparse farmland scene shown in the fourth row of Fig. 5, the models may fail because they cannot extract a sufficient number of keypoints from the UAV and satellite images. Notably, in this farmland case, matching points cluster primarily in regions with slight texture variations rather than uniform fields. This suggests that the model still relies on local repeatable features. A more detailed discussion is provided in Section VI-D.

We visualize the distance errors for a specific flight segment in the second AerialVL scene (see Fig. 8). In this segment, as the scene becomes progressively less texture-rich [Fig. 8(a)–(d)], both SP + SG and XFeat + LG show noticeable performance degradation. In contrast, our method is more robust to texture changes, exhibiting fewer breakpoints. When the scene changes to the case shown in Fig. 8(d), visual methods may not provide sufficiently reliable predictions.

TABLE IV
PERFORMANCE COMPARISON UNDER DIFFERENT TRAINING STRATEGIES

Method	Strategy	MMA@10	MHA@10
XFeat*	default	91.6	82.2
	SP (keypoint)	85.1	75.3
	w/o Synthetic data	88.3	79.7
XFeat	zero-shot	78.7	60.3
	self-supervised	79.3	60.7
SP	zero-shot	80.6	72.7
SP*	default	92.2	89.1
DLK*	default	–	73.6
	w/ Synthetic data	–	73.4

C. Effectiveness Analysis

We analyze the impact of different strategies on performance using the texture-rich subset of the Co-Lo test set.

Ablation Study on XFeat*: 1) using SP as the keypoint detector paired with the XFeat* descriptor; 2) removing the COCO20k synthetic data from the training pipeline; 3) optimizing the descriptor using keypoint correspondences predicted by XFeat itself (self-supervised); and 4) replacing the XFeat with a pretrained SP, while using the same pipeline to construct sparse keypoint pairs for fine-tuning (SP*). The results are shown in Table IV.

In experiment 1, SP tends to favor corner-like structures and detects fewer keypoints than XFeat*, which reduces perfor-

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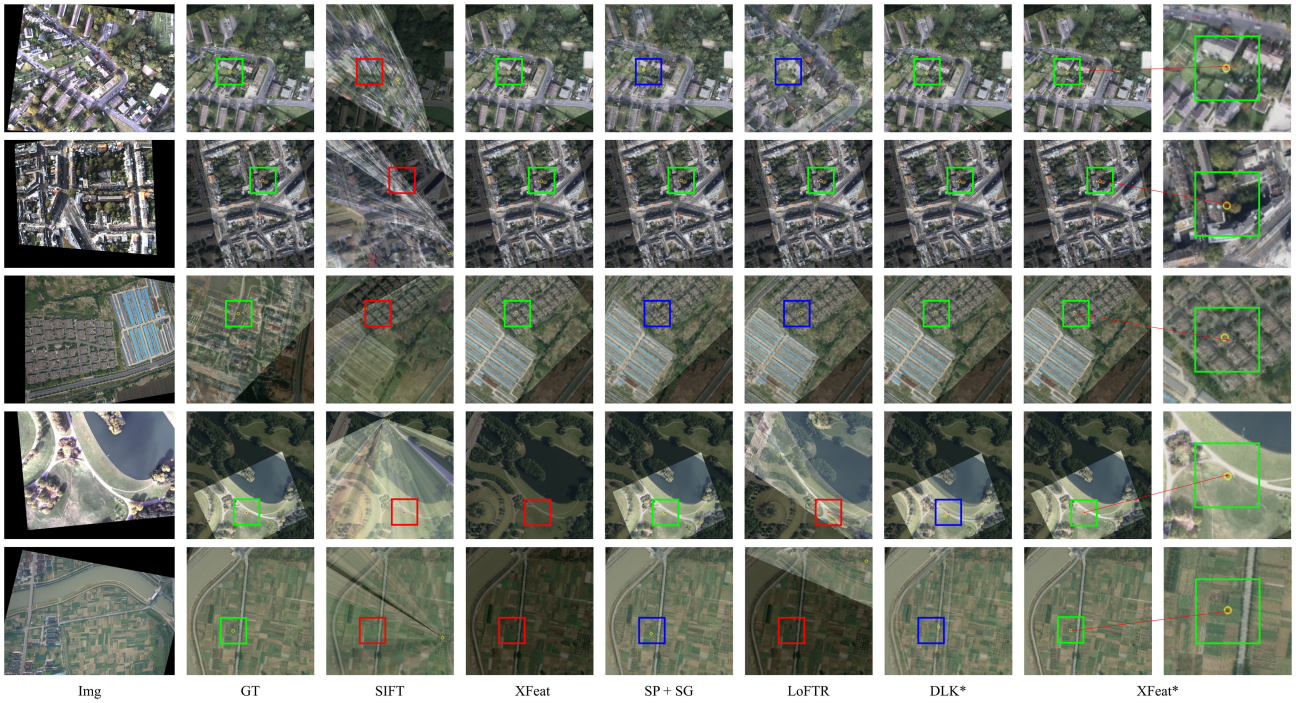


Fig. 7. Visualization of UAV images warped into the satellite coordinate system.

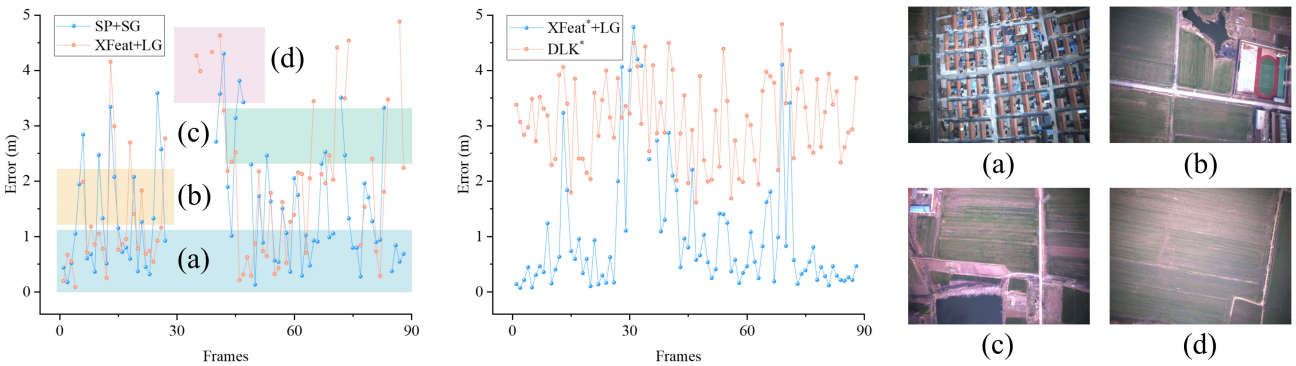


Fig. 8. Distance errors for the second Co-Lo (AerialVL) sequence; texture becomes progressively sparser from (a) to (d).

mance under our evaluation metrics. Nevertheless, compared with the original SP + SG baseline, this configuration still yielded a 4.5 pp performance improvement on MMA@10. This suggests that the RoMa-generated correspondences can provide effective supervision for descriptor learning, even when using a different detector.

In experiment 2, COCO20k was removed from the training pipeline. We observed a decrease of 3.3 pp in MMA@10 and 2.5 pp in MHA@10. These results indicate that even for cross-domain matching, additional natural-image data is beneficial for improving descriptor robustness.

In experiment 3, the self-supervised setting brings limited improvement over zero-shot XFeat. One possible reason is that self-supervision may improve generalization, but it can also introduce noise into the positive samples. Similarly, SP uses synthetic homographic transformations as geometric constraints. In contrast, in UAV–satellite image pairs, accurate reprojection relationships and a strict homography model are generally unavailable. On the other hand, if self-supervision relies only on high-confidence correspondences, many potentially useful regions may be ignored. By comparison, our method filters dense correspondences using thresholding, homography consistency, and sparse keypoint constraints, producing more reliable point pairs. The experimental results indicate that this strategy is more suitable for cross-domain descriptor optimization.

Experiment 4 was designed to examine whether the sparse correspondences constructed from dense correspondences are biased toward specific models. For example, SP tends to focus on corner-like structures, whereas ALIKE tends to respond to locally salient regions. These types of feature correspondences should all be covered by the dense correspondences. The results show that, compared with zero-shot SP, SP* improves MMA@10 and MHA@10 by 11.6 and 16.4, respectively. This trend is consistent with that observed for XFeat*, indicating that the proposed weakly supervised cross-domain adaptation

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strategy is not limited to a single model. If this strategy is applied to a stronger sparse feature model, the matching performance may be further improved.

Control Experiment on DLK:* We trained DLK* with additional synthetic supervision constructed from multitemporal satellite–satellite image pairs. This setting yields no clear improvement, and some metrics degrade slightly. We suggest that multitemporal satellite pairs are insufficient to characterize the complex domain gap between UAV viewpoints and satellite images. Consequently, this synthetic data strategy was excluded from our final training pipeline.

D. Limitations and Future Work

Although the weakly supervised cross-domain adaptation strategy improves descriptor training by establishing keypoint correspondences, several limitations remain.

Correspondence Noise: When using the sampled RoMa correspondences as pseudo-labels, the average reprojection error is approximately 4 px. In texture-sparse scenes, the original XFeat detects fewer keypoints. We aim to obtain more keypoints. However, biased pseudo-labels may introduce incorrect supervision for keypoint locations.

In this work, we freeze the keypoint detection branch of XFeat. Our goal is to treat the sampled RoMa correspondences as pseudo-labels and train the descriptor in a teacher-student manner to mimic RoMa's correspondence behavior. Under this setting, the correspondence error mainly manifests as noise in descriptor supervision. XFeat descriptors typically aggregate local cues within an 8×8 neighborhood. As long as the reprojection error (4 px) stays within this range, the descriptor supervision still covers the dominant discriminative region. The impact on descriptor learning is typically limited.

In fact, bias propagation caused by pseudo-labels is a common challenge in weakly supervised or self-supervised approaches. In future work, we aim to establish more cross-domain keypoint correspondences for stable matching. Reducing the reprojection error will be crucial for enabling joint training of the keypoint detector and descriptor.

Practical Applications: Although DLK is robust in many texture-sparse environments, it may still fail in regions with almost no distinguishable features, such as deserts and open water, or in areas with severe terrain height variations. For real-world deployment, we will explore a more resilient localization framework by fusing visual cues with inertial measurements (INS/IMU) to improve reliability in extremely low-texture scenarios.

VII. Conclusion

We investigate UAV–satellite image registration and continuous localization in scenarios where no camera-pose-based reprojection can be established between UAV and satellite images, and where the scene does not strictly satisfy a homography model. To address these issues, we construct Co-Lo, a calibrated UAV–satellite dataset that supports both pairwise registration evaluation and continuous localization evaluation. Based on this dataset, we introduce a dual evaluation protocol that separates algorithmic matching performance from sequential localization stability, enabling a more rigorous analysis of cross-domain matching methods.

We further propose a weakly supervised cross-domain adaptation strategy that transfers dense correspondences into supervisory signals for sparse feature learning. Specifically, dense matches are filtered and converted into sparse keypoint correspondences for descriptor adaptation, while pseudo-homography matrices are used to supervise DLK optimization. This design allows descriptor learning and homography refinement to benefit from cross-domain correspondence cues without requiring manual pixel-level annotations.

Experimental results show that the adapted models improve matching quality. In particular, XFeat* outperforms its zero-shot counterpart in both matching accuracy and homography accuracy, and also improves continuous localization performance. Additional experiments with SP* further show that the proposed adaptation strategy is not limited to a single sparse feature model. In texture-sparse scenes, DLK* provides competitive image-level alignment performance, indicating that learned feature-map alignment remains effective when reliable sparse correspondences are difficult to obtain.

Appendix A
Ground-Truth Homography Construction

During practical data processing, we observed that in-plane rotations of UAV images can substantially degrade the matching performance of RoMa [40]. To mitigate this issue, we first rotate each UAV image to align it with true north using the UAV's recorded yaw angle. If the yaw angle is unavailable, the rotation angle is provided manually. The corresponding rotation transform $H^{\mathrm{N}}_{\mathrm{u}\to\mathrm{u}}$ is defined as

$$H^{\mathrm{N}}_{\mathrm{u}\to\mathrm{u}} = T_{\mathrm{c}} R(\varphi) T_{\mathrm{c}}^{-1}$$
$$T_{\mathrm{c}} = \begin{bmatrix} 1 & 0 & -c_{\mathrm{x}} \\ 0 & 1 & -c_{\mathrm{y}} \\ 0 & 0 & 1 \end{bmatrix}, \; R(\varphi) = \begin{bmatrix} \cos\varphi & -\sin\varphi & 0 \\ \sin\varphi & \cos\varphi & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{A1}$$

Here, $(c_{\mathrm{x}}, c_{\mathrm{y}})$ denotes the image-center pixel coordinate, and φ denotes the yaw angle.

We use RoMa to obtain an initial geometric correspondence $\hat{H}^{\mathrm{roma}}_{\mathrm{u}\to\mathrm{s}}$, and then manually refine it by estimating a corrective homography $\Delta H^{\mathrm{art}}_{\mathrm{u}\to s}$. The homography after rotation is given by

$$H^{\mathrm{rot}}_{\mathrm{u}\to\mathrm{s}} = \Delta H^{\mathrm{art}}_{\mathrm{u}\to\mathrm{s}} \hat{H}^{\mathrm{roma}}_{\mathrm{u}\to\mathrm{s}} \; . \tag{A2}$$

Given the rotation transform $H^{\mathrm{N}}_{\mathrm{u}\to\mathrm{u}}$, the ground-truth homography from the UAV image I_{u} to the satellite image I_{s} is

$$H^{\mathrm{gt}}_{\mathrm{u}\to\mathrm{s}} = H^{\mathrm{rot}}_{\mathrm{u}\to\mathrm{s}} H^{\mathrm{N}}_{\mathrm{u}\to\mathrm{u}} = \Delta H^{\mathrm{art}}_{\mathrm{u}\to\mathrm{s}} \hat{H}^{\mathrm{roma}}_{\mathrm{u}\to\mathrm{s}} \; H^{\mathrm{N}}_{\mathrm{u}\to\mathrm{u}}. \tag{A3}$$

Not all samples are manually calibrated. To reduce annotation cost, we preselect a set of GCPs and compute the reprojection error. Manual fine-tuning is applied only when this error exceeds 3 px. This procedure primarily focuses on challenging cases with sparse textures, where the RoMa method struggles to provide reliable matches. The overall pipeline is illustrated in Fig. 9. The dataset calibration is used only for evaluation.

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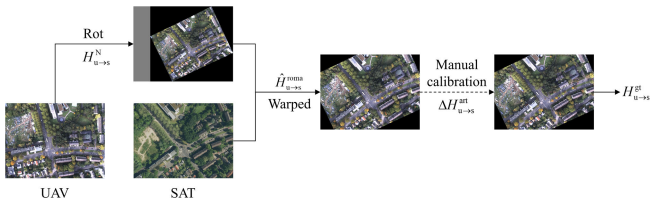


Fig. 9. Flow of Data Annotation. Gray areas are for visualization only.

The above procedure is designed for UAV–satellite registration under texture discrepancies. To accommodate additional viewpoint variations $I_{\mathrm{u}}^{\mathrm{warp}}$, we manually apply a planar projective transformation (e.g., perspective warping and rotation) $\dot{H}_{\mathrm{u}\to\mathrm{u}}^{\mathrm{warp}}$ to the original UAV image I_{u}. Accordingly, the homography $H_{\mathrm{v_u}\to\mathrm{s}}^{\mathrm{gt}}$ from the synthesized image $I_{\mathrm{u}}^{\mathrm{warp}}$ to the satellite image I_{s} can be expressed as

$$\begin{aligned} &\dot{H}_{\mathrm{u}\to\mathrm{u}}^{\mathrm{warp}} : I_{\mathrm{u}} \to I_{\mathrm{u}}^{\mathrm{warp}} \\ &H_{\mathrm{v_u}\to\mathrm{s}}^{\mathrm{gt}} = H_{\mathrm{u}\to\mathrm{s}}^{\mathrm{gt}} \left(\dot{H}_{\mathrm{u}\to\mathrm{u}}^{\mathrm{warp}}\right)^{-1}. \end{aligned} \tag{A4}$$

Appendix B
Evaluation Metric Definitions

In feature matching evaluation, MAE and RMSE are not used. Satellite imagery acquired at different times often exhibits geo-registration offsets, which can lead to inconsistent conclusions when distance-based metrics are used. Therefore, we evaluate algorithm performance using image-matching metrics (MMA/MHA).

To evaluate performance during continuous flights, we select the satellite image that is temporally closest to each flight sequence. This choice reduces systematic bias due to temporal misalignment. We dynamically crop a reference template from the satellite image based on the flight altitude.

During initialization, we provide accurate latitude–longitude priors and constrain the initial heading to the discrete orthogonal direction that is closest to true north (e.g., the four orthogonal candidates at 90° intervals). This constraint mitigates match failures caused by large in-plane rotation discrepancies and helps preserve continuity across sequences.

Assume that sequence s contains images N_s. The localization error for frame i is defined as

$$e_{s,i} = \|\hat{y}_{s,i} - y_{s,i}\|_2 \tag{A5}$$

where $\hat{y}_{s,i}$ and $y_{s,i}$ denote the estimated and ground-truth positions in the Mercator coordinate system, respectively. The interframe displacement $\Delta d_{s,i}$ is the ground-truth travel distance from frame $i-1$ to frame i. For $i = 1$, we set $\Delta d_{s,i} = 0$.

With a success threshold $\tau = 10$ m, we define the per-frame success indicator as

$$1_{s,i} = \begin{cases} 1, & 0 \le e_{s,i} < \tau \\ 0, & \text{otherwise.} \end{cases} \tag{A6}$$

We define continuous successful localization as a maximal contiguous interval $I_{s,k}$:

$$I_{s,k} : \left[a_{s,k}, b_{s,k}\right] \quad \text{s.t.} \quad 1_{s,i} = 1, \quad \forall i \in \left[a_{s,k}, b_{s,k}\right] \tag{A7}$$

where k indexes the kth continuous success segment and $1 \le a_{s,k} < b_{s,k} \le N_s$. The traveled distance within interval $I_{s,k}$ is

$$L_{s,k} = \sum_{i=a_{s,k}}^{b_{s,k}} \Delta d_{s,i}. \tag{A8}$$

The maximum continuous successful-localization distance $L_s^{\max}$ and the sum of successfullocalization distances over all segments L_s^{sum} are defined as

$$\begin{aligned} L_s^{\max} &= \max_k L_{s,k} \\ L_s^{\mathrm{sum}} &= \sum_k L_{s,k} = \sum_{i=1}^{N_s} 1_{s,i} \Delta d_{s,i}. \end{aligned} \tag{A9}$$

The maximum continuous stable localization ratio ($\mathrm{R}_{\max}$) and the successful coverage ratio ($\mathrm{R}_{\mathrm{suc}}$) are defined as

$$\begin{aligned} \mathrm{R}_{\max} &= \frac{\sum_{s=1}^{S} L_s^{\max}}{\sum_{s=1}^{S} D_s} \\ \mathrm{R}_{\mathrm{suc}} &= \frac{\sum_{s=1}^{S} L_s^{\mathrm{sum}}}{\sum_{s=1}^{S} D_s}. \end{aligned} \tag{A10}$$

Here, D_s denotes the total traveled distance of sequence s, S is the number of sequences.

Appendix C
IC-LK Optimization Details

The classical Lucas–Kanade (LK [15]) method aligns an input image $I(\mathbf{x})$ to a template $T(\mathbf{x})$ by optimizing the warp parameters $\mathbf{p}$ to minimize the sum of squared differences (SSDs):

$$E(p) = \sum_x \|I(W(\mathbf{x};\mathbf{p})) - T(\mathbf{x})\|_2^2 \tag{A11}$$

where $W(\mathbf{x};\mathbf{p})$ denotes the warp and $\mathbf{p} = (\mathrm{p}_1, \mathrm{p}_2, \ldots, \mathrm{p}_n)^{\mathrm{T}}$ is the parameter vector. We consider a homography warp ($n = 8$). At iteration k, given the current parameters $\mathbf{p}_k$, we estimate an increment $\Delta\mathbf{p}$ and update the parameters as

$\mathbf{p}_{k+1} = \mathbf{p}_k + \Delta\mathbf{p}$. The incremental objective is

$$E(\Delta\mathbf{p}) = \sum_{\mathbf{x}} \|I\left(W\left(\mathbf{x};\mathbf{p}_k + \Delta\mathbf{p}\right)\right) - T(\mathbf{x})\|_2^2. \tag{A12}$$

Using a first-order Taylor expansion around $\mathbf{p}_k$ yields

$$E(\Delta\mathbf{p}) \approx \sum_{\mathbf{x}} \left\| I\left(W\left(\mathbf{x};\mathbf{p}_k\right)\right) + \nabla I \frac{\partial W}{\partial \mathbf{p}}\bigg|_{\mathbf{p}_k} \Delta\mathbf{p} - T(\mathbf{x}) \right\|_2^2 \tag{A13}$$

where $\nabla I = (\partial I/\partial x, \partial I/\partial y)$ denotes the image gradient. Since both ∇I and $\partial W/\partial\mathbf{p}$ depend on $\mathbf{p}_k$, they must be recomputed after each update.

To improve efficiency, the IC-LK algorithm applies the increment warp $\Delta\mathbf{p}$ to the template and rewrites the objective as

$$E(\Delta\mathbf{p}) = \sum_{\mathbf{x}} \|T(W(\mathbf{x};\Delta\mathbf{p})) - I\left(W\left(\mathbf{x};\mathbf{p}_k\right)\right)\|_2^2. \tag{A14}$$

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TABLE V
COMPARISON OF MATCHING AND REGISTRATION METHODS ON TEXTURE-RICH SCENES IN CO-LO. RESULTS ARE REPORTED IN MMA AND MHA AT THRESHOLDS @3/@5/@7/@10 (HIGHER IS BETTER). THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	MMA				MHA			
	@3	@5	@7	@10	@3	@5	@7	@10
Traditional method								
SIFT	6.0	7.0	7.3	7.9	0.8	1.0	1.8	2.0
Deep-learning detector-based methods								
SP + SG	64.9	77.3	79.4	80.6	41.3	51.9	67.1	72.7
ALIKED + LG	66.0	79.3	81.5	82.5	41.8	53.4	68.6	74.7
XFeat + LG	56.8	73.0	76.9	78.7	27.3	44.6	52.9	60.3
Deep-learning detector-free methods								
LoFTR	57.3	68.4	70.8	72.2	32.3	51.6	58.4	62.9
ASpanFormer	62.0	72.2	74.5	75.9	39.6	53.9	59.6	63.9
Ours								
XFeat* + LG	**72.4**	**85.3**	**87.6**	**91.6**	**60.1**	**69.4**	**75.1**	**82.2**
DLK*	–	–	–	–	41.2	51.2	68.4	73.6

A first-order Taylor expansion of $T(W(\mathbf{x};\Delta\mathbf{p}))$ around $\Delta\mathbf{p}=0$ yields

$$E(\Delta\mathbf{p}) \approx \sum_{\mathbf{x}} \left\| T(\mathbf{x};\mathbf{0}) + \nabla T \frac{\partial W}{\partial \mathbf{p}}\bigg|_{\mathbf{p}=0} \Delta\mathbf{p} - I\left(W\left(\mathbf{x};\mathbf{p}_k\right)\right) \right\|_2^2. \tag{A15}$$

Since $\partial w/\partial \mathbf{p}|_{\mathbf{p}=0}$ is evaluated at the identity warp, it remains constant across iterations and can be precomputed. Following [14], the Gauss-Newton update is

$$\Delta\mathbf{p} = \left(\mathbf{J}^{\mathrm{T}}\mathbf{J}\right)^{-1}\mathbf{J}^{\mathrm{T}}\mathbf{r} \tag{A16}$$

where $\mathbf{J} = \nabla T \partial W/\partial \mathbf{p}|_{\mathbf{p}=0}$ and $\mathbf{r}(\mathbf{x}) = I(W(\mathbf{x};\mathbf{p}_k)) - T(\mathbf{x})$. The warp is then updated via inverse composition [see Fig. 4(a) and (b)]:

$$W\left(\mathbf{x},\mathbf{p}_{k+1}\right) \leftarrow W\left(\mathbf{x},\mathbf{p}_k\right) \circ W\left(\mathbf{x},\Delta\mathbf{p}_k\right)^{-1}. \tag{A17}$$

For homography estimation, the warp $W(\mathbf{x};\mathbf{p})$ maps a template coordinate $\mathbf{x} = [x,y]^{\mathrm{T}}$ to an image pixel $\mathbf{x}' = [x',y']^{\mathrm{T}}$ and can be parameterized as

$$W(\mathbf{x};\mathbf{p}) = \frac{1}{1+p_7x+p_8y}\begin{bmatrix}(1+p_1)x+p_2y+p_3\\ p_4x+(1+p_5)y+p_6\end{bmatrix}. \tag{A18}$$

The Jacobian at the identity warp is

$$\frac{\partial W}{\partial \mathbf{p}}(\mathbf{x};\mathbf{0}) = \begin{bmatrix} x & y & 1 & 0 & 0 & 0 & -x^2 & -xy \\ 0 & 0 & 0 & x & y & 1 & -xy & -y^2 \end{bmatrix}. \tag{A19}$$

After convergence, the homography can be written as

$$H_{\mathrm{p}} = \begin{bmatrix} 1+p_1 & p_2 & p_3 \\ p_4 & 1+p_5 & p_6 \\ p_7 & p_8 & 1 \end{bmatrix}. \tag{A20}$$

APPENDIX D
ALGORITHMIC EVALUATION IN TEXTURE-RICH SCENARIOS

As shown in Table V, XFeat*, trained on cross-domain keypoint correspondences, achieves 91.6 MMA@10 and 82.2 MHA@10, outperforming the original XFeat [41] by 16.4% and 36.3%, respectively. For DLK*, the initial parameters $\mathbf{p}$ are initialized using SP [29] + SG [30]. Compared with sparse-texture scenes, the performance does not improve further. In texture-rich scenes, keypoint matching produces stable correspondences and accurate alignment. As a result, under strict error thresholds, keypoint-based methods often outperform DLK-based approaches.

Even in texture-rich scenes, SIFT [46] still fails to achieve satisfactory performance. This poor performance is primarily because classical pipelines rely on local grayscale similarity around keypoints. However, satellite images and raw UAV images often exhibit substantial appearance discrepancies in texture, which violates this assumption (see Fig. 10).

APPENDIX E
SEQUENCE EVALUATION ON CO-LO (DENSEUAV)

We report the results of several methods on the Co-Lo (DenseUAV) dataset (see Table VI). Since DenseUAV provides high-precision geographic coordinates and maintains a strictly nadir view, no additional manual calibration was performed. Compared with the results on Co-Lo (AerialVL), all methods show higher errors in MAE, RMSE, and R_{max} . This increase is primarily due to the characteristics of DenseUAV [21], which consists of UAV sequences captured below 100 m and features densely packed high-rise buildings. This creates pronounced perspective differences between UAV and satellite views. The planar homography assumption can introduce considerable geometric errors,

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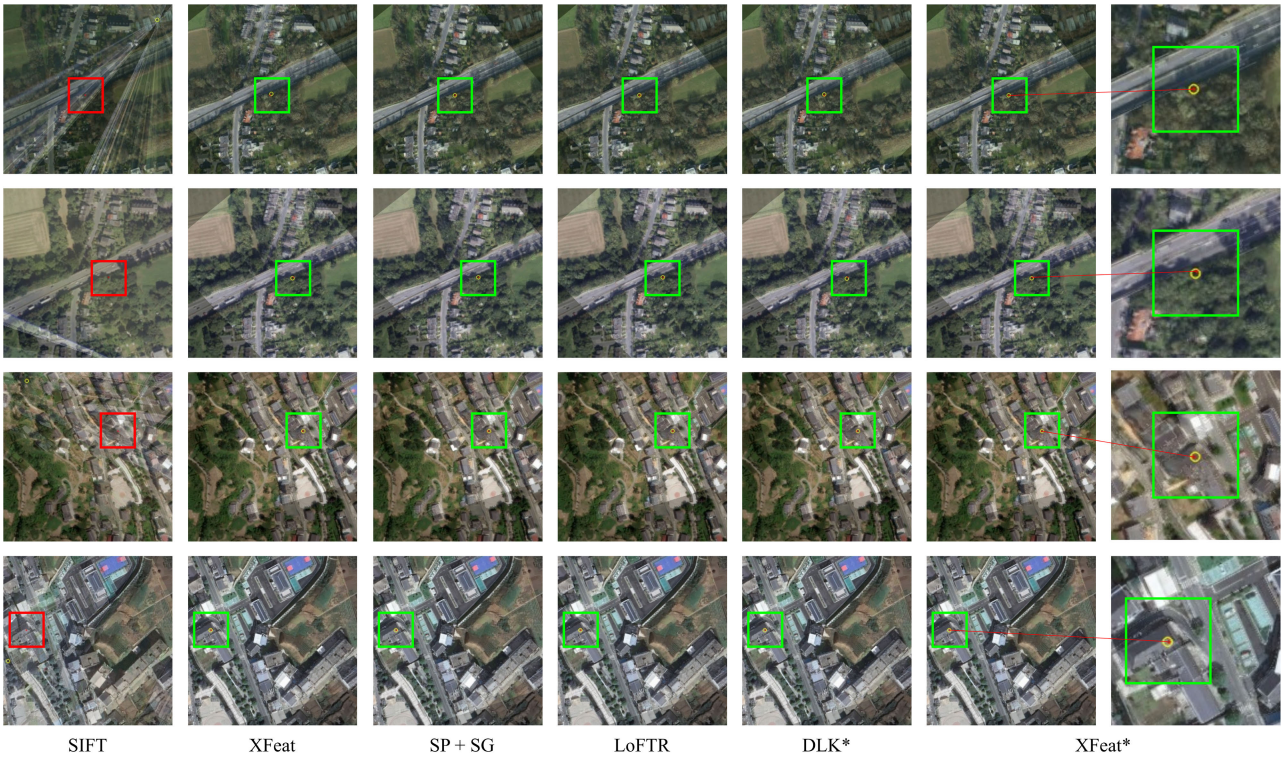


Fig. 10. Qualitative visualizations of several methods. The warped UAV image is overlaid on the satellite image. Red dots indicate the ground-truth locations, and yellow circles indicate the predicted locations. Red boxes denote failed predictions, whereas green boxes denote successful predictions.

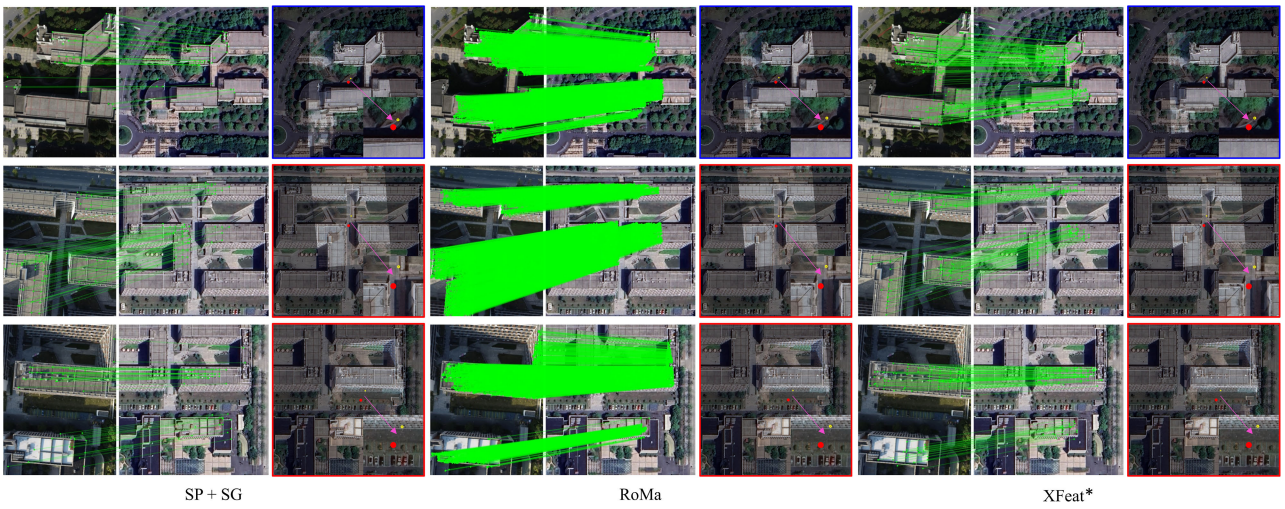


Fig. 11. Qualitative results on the Co-Lo (DenseUAV) dataset. Blue boxes indicate predictions with localization error below 10 m, while red boxes indicate errors of 10 m or more.

making this dataset particularly challenging. The gains of XFeat* mainly stem from a substantial increase in the number of inliers, which enables more stable and reliable matching.

Qualitative results are presented in Fig. 11. On this dataset, detector-based methods typically achieve an average localization error of about 4.35 m. Their inlier correspondences are often dominated by building regions with strong perspective effects. This causes the estimated homography to deviate from the ground plane. Without explicitly constraining the ground as the reference plane, these methods generally cannot achieve lower localization errors. In addition, we do not report results for DLK because feature matching fails to provide an initial estimate of $\mathbf{p}$.

The DenseUAV dataset has an average sampling interval of approximately 20 m. As a result, retrievalbased approaches produce localization errors that are significantly larger than matching-based errors, which are approximately $4-5$ m. Under the severe viewpoint differences caused by low-altitude flights, reducing the localization error without introducing digital orthophotos (DOPs) or DSMs requires further investigation.

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TABLE VI

COMPARISON OF CONTINUOUS LOCALIZATION METHODS IN CO-LO (DENSEUAV) DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Method	MAE	RMSE	R_{max}	R_{suc}	Inliers
Detector-based methods					
SP + SG	4.31	4.82	44.9	94.3	63
ALIKED + LG	4.48	4.98	47.8	94.4	91
XFeat + LG	4.34	4.83	40.1	93.9	109
Detector-free methods					
LoFTR	**3.99**	**4.48**	46.6	95	244
ASpanFormer	4.31	4.82	50.1	95	400
Dense matching methods					
RoMa	4.10	4.60	**64.2**	**96.4**	**1679**
Ours					
XFeat* + LG	4.25	4.75	49.2	94.7	203

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